Technical review

Item #2 · 12-mailbox.mbox#3@2699 · Run ID: 20261113T232800Z-08b8d1

Path	`R:\evidence\mail\12-mailbox.mbox#3@2699`
Size (bytes)	490
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.mbox
Detected type (signature)	Email message in an mbox (RFC 5322)
Status	Needs attention
Derived from (mbox evidence)	Message #3 of the mbox R:\evidence\mail\12-mailbox.mbox (md5 f4e8e05ccc50d543ed3703f21cb965d4, sha256 2e39de637dad0f16469632facf7544dfe4fc0d39bf7210dd579b7a74ba0f5ebe), stored from byte offset 2748, 490 bytes (separator line at offset 2699). The custody of these bytes is the custody of that mbox file.
Case	RV-SAMPLE-001
Evidence ID	E0016
Case registry	
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`9d13579ab0f6860e938f33c35b485fca29c2417dd5b6d57fa59c892eb6a4499a`
md5	`c00d9f9edc005b7e74a14f28fbad963e`

Findings

[Low] No DKIM signature (DKIM_NOT_SIGNED)

What it shows	The message carries no DKIM-Signature header.
What it does not prove	Many legitimate servers do not sign, and a missing signature is common; it proves nothing about authenticity either way.

[Medium] Received times go backwards (RECEIVED_TIME_NOT_MONOTONIC)

What it shows	Hop 2 was stamped 2026-11-11T09:40:00Z, 1200 s before the previous hop 1 (2026-11-11T10:00:00Z), beyond the tolerated clock difference (1 such step(s)).
What it does not prove	Server clocks are sometimes wrong and some servers add Received headers out of order; it does not show that a header was inserted or edited.

[Medium] Date header later than the first server (DATE_AFTER_FIRST_HOP)

What it shows	The Date header (2026-11-11T11:30:00Z) is 5400 s later than the time the first server received the message (2026-11-11T10:00:00Z).
What it does not prove	The sender's clock may be wrong; it does not show that the date was changed after sending.

[Low] Time zone inconsistent (TIMEZONE_INCONSISTENT)

What it shows	The Date header uses zone +0500 while the sender organisation's first server used +0000 (sender_server_zone_differs).
What it does not prove	Clients and servers are often set to different zones, travellers keep their home zone; it does not show that the date was edited.

[Low] Message-ID from another domain (MESSAGE_ID_DOMAIN_MISMATCH)

What it shows	The Message-ID was generated under mailer.example.com, not under the From domain example.org.
What it does not prove	Mail clients, sending services and webmail hosts generate Message-IDs with their own names; it does not show that the message is forged.

Email analysis

Message in mbox	#3 @2699 (490 B): From bruno@example.org Wed Nov 11 09:40:00 2026
Message-ID	x33@mailer.example.com
From	Bruno Proveedor <bruno@example.org>
Subject	Albarán 33
Date header	Wed, 11 Nov 2026 16:30:00 +0500 (2026-11-11T11:30:00Z)
DKIM mode / key file	offline · R:\inputs\dkim-keys.txt · 4 key(s)

Received hops (first server = 1)

Hop 1	From host: [192.0.2.30] · Received by: smtp.example.org · Date stamped: 2026-11-11T10:00:00Z
Hop 2	From host: smtp.example.org · Received by: mx.acme.example · Date stamped: 2026-11-11T09:40:00Z

DKIM signatures (offline verification)

ARC chain	none (0)

Files written by the analysis

headers.json (headers)	md5 8f24c28cdb300afe68646772d06eb0e2 · sha256 7e8bd2ff7d8863f93b44a808d7e4c8063664a88d042af873cef5f96bca3b14e7
dkim.json (dkim)	md5 89c8df517c799bdef19769cc6069a780 · sha256 bdd96dbaa00537c61cc388dc305d9fecec69ae98ef74516ff0c2c313b86fccda

Method

Each file was read once to compute its digests, then analysed as email: .eml as an RFC 5322 message, .msg through its MAPI properties, and each message of an mbox as an item of its own (each in a separate worker process with time and memory limits when worker isolation is on, as recorded in the audit log). The header section, Received hops and dates were read and compared; DKIM and ARC signatures were verified offline with the public keys the examiner supplied (no DNS query unless online mode was chosen); Authentication-Results were recorded as stated evidence; attachments were hashed; and BEC heuristics compared the addresses with the organisation's own domains. Every step and every file written is recorded in the hash-chained audit log.

Audit trail

Audit entries	#31 – #36
Log folder	`R:\case\runs\rv-mailauth\20261113T232800Z-08b8d1\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- DKIM results depend on the public keys supplied: a check is only meaningful with the key that was valid when the

message was sent. Keys are rotated, so a current key can fail a genuine message and a missing key leaves the signature unchecked.

- A missing DKIM signature is common and proves nothing; a valid signature covers only the signed header fields and the body, not who wrote the message.
- Outlook .msg files do not keep the original message: the body is re-encoded (so DKIM cannot be checked) and the original header section is kept only when transport headers were stored.
- Lookalike-domain, Reply-To, display-name and payment-wording checks are heuristics with false positives and false negatives; the organisational domain comes from the bundled Public Suffix List, whose version is recorded in environment.json and may differ from the list in force when the message was sent.
- Received, Date and Authentication-Results headers are text that servers and clients wrote; unsigned headers can be added or edited by anyone who handled the file.
- In an mbox, body lines starting with "From " are escaped with '>'; un-escaping follows the selected variant (mboxrd by default) and cannot always restore the original line, which can affect DKIM body hashes.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.